Junrong Chen

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Education

Doctor of Philosophy in Computer Science, University of California, Irvine *Sept. 2026 –Present*
Advisor: Prof. Judy Hoffman
Bachelor of Science, University of California, San Diego *Sept. 2022 – Dec. 2025*
Double Major: Mathematics-Computer Science; Cognitive Science (Specialization: Machine Learning and Neural Computing); **GPA:** 3.845/4.00

Research Interests

World Models, Embodied AI, Machine Learning, Artificial Intelligence

Publications and Conference Presentations

1 Yidong Jiang, **Junrong Chen**, Eftychia Makri, Jialin Chen, Peiwen Li, Ali Maatouk, Leandros Tassiulas, Eliot Brenner, Bing Xiang, Rex Ying
Fin-RATE: A Real-world Financial Analytics and Tracking Evaluation Benchmark for LLMs on SEC Filings PDF
Submitted to SIGKDD 2026 and under review.

2. Qiyue Gao*, Xinyu Pi*, Kevin Liu, **Junrong Chen**, Ruolan Yang, Xinqi Huang, Xinyu Fang, Lu Sun, Gautham Kishore, Bo Ai, Stone Tao, Mengyang Liu, Jiaxi Yang, Chao-Jung Lai, Chuanyang Jin, Jiannan Xiang, Benhao Huang, David Danks, Hao Su, Tianmin Shu, Ziqiao Ma, Lianhui Qin, Zhiting Hu
Do Vision-Language Models Have an Internal World Model? Towards an Atomic Evaluation PDF
Findings of 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)
World Models: Understanding, Modeling, Scaling, and Applications workshop at ICLR 2025 (Poster)

3. PAN Team Institute of Foundation Models (contributing author: **Junrong Chen**).
PAN: A World Model for General, Interactable, and Long-Horizon World Simulation PDF
arXiv:2511.09057, 2025.

4. **Junrong Chen**, Jan Kwong, Sarah Paull, Sarah Creel
Whisper Automatic Speech Recognition and humans are similarly sensitive to accents, but use sentence context differently PDF
Submitted to Journal and under review.

5. Sarah Creel, **Junrong Chen**, Caroline Hall-Sherr, Sadrac Santacruz Ibarra, Carly Villongco, Kaitlyn Chou
Whispers of knowledge: ASR vs human performance in child transcription
Abstract accepted at the ISCA/ITG LLMDiversity 2025.

6. **Junrong Chen**, Jan Kwong, Sarah Creel
Accented sentence and word recognition: Humans versus Whisper automatic speech recognition
Abstract accepted for oral presentation at the 187th Meeting of the Acoustical Society of America (187th ASA) Link
GenAI Summit UCSD (Poster), Feb.2025

Research Experience

UCSD *Mix lab* | Research Assistant
Mentor: Professor Zhiting Hu

Project 1: World Model Evaluation Link (TR Released, Eval MS in Prep) *Mar.2025-Present ·14mos*

- Fine-tuned and evaluated PAN-2, the second-generation PAN world model, across domains including robotics, autonomous driving, and gaming.
- Led the robotics track and initiated the autonomous driving thread, responsible for dataset streaming across 6+ sources, model training and fine-tuning, and developing 20+ demos and pipelines, as well as 8+ controlled experiments, including IDM model training and use of the SimplerEnv simulator to validate key model capabilities.
- Skills: Python, Hugging Face, PyTorch, CUDA, Slurm, Bash

Project 2: World Model Benchmark Link *Mar.2024-Jun.2025 ·16mos*

- Developed a fine-grained evaluation framework (23 dimensions) across 6 simulated environments with controlled counterfactual simulations, testing 11 commercial/open-source VLMs through 517 controlled experiments.
- Led data pipeline development and collected data using Habitat-Lab and ManiSkill2, ThreeDWorld simulators for over 6 datasets with 12k+ cases in total. Utilized 11+ open-source datasets to evaluate models, 5 commercial VLMs and 6 open-source VLMs. Conducted data analysis, visualization, and drafted conclusions.
- Drafted the Introduction, Experimental Design, and Appendix; Produced a promotional video and website.

- Skills: Python, Hugging Face, PyTorch, CUDA, Docker

Yale University *Graph and Geometric Learning Lab* | Research Assistant. *Sept.2025 – Present · 7 mos*

Mentor: Professor Rex (Zhitao) Ying

- Co-authored Fin-RATE, a benchmark for evaluating LAGs on financial reasoning over SEC filings, spanning detail-oriented reasoning, cross-company comparison, and longitudinal tracking across 7,500 QA instances derived from 2,472 filings covering 43 companies and 36 industries.
- Designed the evaluation framework, error taxonomy, and human validation protocol; conducted controlled experiments characterizing model degradation across task types and analyzed corresponding failure patterns.
- Performed data analysis and visualization, drafted the main results section, and contributed to revisions throughout the manuscript.
- Skills: Python, Hugging Face, SEC filings analysis

Cornell University *Jennifer J. Sun's lab* | Undergraduate Researcher *Mar.2024- Mar.2025 ·13mos*

Mentor: Professor Jennifer J. Sun

- Developed an automated pipeline to enforce programmatic cycle consistency in text-to-image models using LLM-generated verification code and visual grounding tools (Workshop manuscript in preparation).
- Created a dataset of 20,000+ images using Stable Diffusion and conducted over 20,000 experiments on models including CLIP, SigLIP, Llava, PaliGemma, and Llama.
- Conduct controlled experiments on how improved prompt fidelity and image accuracy compared to baseline generation pipelines (e.g., Stable Diffusion).
- Skills: Python, Hugging Face, PyTorch

UCSD *Language Acquisition and Sound Recognition Lab* | Lead Researcher *Sept.2023- Sept.2025·25mos*

Mentor: Professor Sarah Creel

Project: Enhancing Speech Recognition using Machine Learning Tools

- Investigated AI bias in speech recognition tools (e.g., Montreal Forced Aligner, Char-siu, Whisper) for diverse speech types-standard American English, accented speech (Mandarin, Spanish), child speech, and distorted speech (noise-vocoded, sine-wave)-to improve inclusivity and accuracy.
- Processed 40,000+ audio files using Whisper for varying model sizes, analyzed data using R, and wrote paper.
- Skills: R, Hugging Face, Python, PyTorch

UCSD *Language Acquisition and Sound Recognition Lab* | Research Assistant *Apr.-Sept. 2023 ·6mos*

Mentor: Professor Sarah Creel

Project: KidTalk and L2Talk

- Investigated the gap between monolingual children's language output and language perception (KidTalk), and the gap between bilingual adults' language output and language perception (L2Talk).
- Collected and analyzed 72 pieces of 3–5-year-old child word-naming, free speech, Goldman-Fristoe Test of Articulation (GFTA) data and 5 pairs of adult data of word-naming data, contributed to experiment design.
- Skills: Matlab, Phon, Praat, Eyelink100, R

UCSD *System Neuroscience Lab* | Research Assistant *Jun. 2023-Mar. 2024 ·10mos*

Mentor: Professor Douglas A. Nitz

- Conducted rat experiment analyzing visual stimuli transitions from overhead to an eye-level perspective.
- Processed 200+ minutes of animal behavior videos using DeepLabCut and SLEAP, Trained machine learning models to track rat positions; Assisted in brain-computer interaction (BCI) drive-building sessions for rats.
- Skills: Python, DeepLabCut, SLEAP

Professional Experience

Ask Margot *(UCLA AI Studio Fellow), Los Angeles, CA* *Aug.2024-Apr.2025 ·9mos*

- Built a web chatbot using AWS Lex and integrated AWS Lambda and SageMaker for sentiment analysis and model training.

Brain machine interface workshop & hackathon*, UCSD* *Sept. 2023 ·1mos*

Mentor: Professor Vikash Gilja (UCSD), Cindy Chestek (UMich), Gal Mishne (UCSD)

- Analyzed neural data with tools (NumPy, Pandas) and machine learning (scikit-learn, PyTorch) to prototype neural prostheses and present ZebraFinch Dataset findings.

TalkMeUp Software Engineering Internship*, Pittsburgh, PA* *Aug.-Sept. 2023 ·2mos*

- Developed and optimized machine learning software solutions to automatically score interview videos; implemented product-focused AI solutions using computer vision and NLP.

Honor and Awards

UCSD Triton Research & Experiential Learning Scholarship (TRELS), Spring Quarter 2024 *Link*

UCSD Provost Honors (2022-2025)

IEEE-Eta Kappa Nu (IEEE-HKN) (2024)

Instructional Apprentice

Undergraduate Teaching Assistant at UC San Diego

Lead discussion sections, hold office hours, answer students' questions via email, grade homework and exams, and proctor exams.

COGS14A - Intro. to Research Methods (Spring Quarter 2024, Professor Steven Barrera)

COGS14B - Intro. to Statistical Analysis (Winter Quarter 2024, Professor Drew Ellen Walker)

Last updated: Apr. 6, 2026